

MetAgent case study

compound_only_enrich_mammalian_RAMP_P_000000398_seed6

Input metabolite set -> pathway evidence -> final metabolic interpretation

Input

Differential metabolites (n=10)

myo-Inositol C6H12O6	C00137
Sucrose C12H22O11	C00089
D-Glucose C6H12O6	C00221
Galactitol C6H14O6	C01697
Galactinol C12H22O11	C01235
Glycerol C3H8O3	C00116

Other differential metabolites

L-Cysteine (C00097), S-Adenosylhomocysteine (C00021), Fumaric acid (C00122), 5,6-Epoxy-8,11,14-eicosatrienoic acid (C14768)

Top pathway

Galactosemia

SMPDB SMP00182 | FDR=2.20e-15
fold-enrichment=1080.8 | matched 6/10

Final verifier totals

supported 18 | unsupported 1 | contradicted 1
unverifiable_v0 20

Final MetAgent report

Verifier-colored complete narrative

supported

unsupported

contradicted

unverifiable_v0

The RaMP enrichment analysis reveals an overwhelming signal for Galactose Metabolism (KEGG map00052; FDR=0.0, fold-enrichment ~ 1,046) and the related disorder module Galactosemia (SMP00182; FDR=0.0, fold-enrichment ~ 1,081). Six of the ten input metabolites map to these entries: myo-inositol (C00137), sucrose (C00089), D-glucose (C00221), galactitol (C01697), galactinol (C01235), and glycerol (C00116).

Membership of sucrose and galactinol in Galactose Metabolism was independently confirmed via pathway membership evidence.

Secondary hits include broader biochemical routes - notably amino sugar and nucleotide sugar metabolism - consistent with galactose and sulfur-amino-acid metabolism intersecting at central carbon intermediates.

Key driver compounds.

Sucrose (C00089) is a canonical entry point for carbohydrate assimilation, classified in Starch and Sucrose Metabolism as well as Galactose Metabolism [retracted: the specific claim that "sucrose is hydrolysed in the Leloir pathway" was not supported by RaMP/KEGG evidence].

The KEGG BFS confirms a forward conversion chain from sucrose to D-glucose (4 hops).

Galactitol (C01697) forms via aldose reductase when galactose accumulates, and galactinol (C01235) is the galactose donor for raffinose-family oligosaccharide synthesis - both are characteristic biomarkers of impaired galactose clearance.

myo-Inositol (C00137) shares downstream connectivity with galactose via glucose-6-phosphate intermediates, providing a plausible biological basis for its co-enrichment in Galactose Metabolism.

Upstream/downstream relationships.

L-Cysteine (C00097) and S-adenosylhomocysteine (C00021) are annotated together in the cysteine and methionine metabolism subsystem, connected via a four-hop bidirectional path.

Fumarate (C00122) appears as a downstream node of the TCA cycle and as a product of SAH metabolism, suggesting that the sulfur amino acid and energy metabolism arms of this dataset converge at fumarate.

Biological significance.

A coordinated elevation or depletion of the six galactose-axis metabolites is characteristic of perturbations in Galactose Metabolism - whether genetic (classic galactosemia variants in GALT) or adaptive.

Altered glycerol and myo-inositol levels may reflect osmotic compensation and inositol lipid signaling stress, respectively.

The concurrent sulfur-amino-acid dysregulation (cysteine ↔ SAH) points to broader metabolic strain in the cysteine and methionine metabolism subsystem.

The literature also links galactose metabolism with amino sugar and nucleotide sugar metabolism (PMID 40325042), consistent with the secondary enrichment observed here.

Limitations.

Causal attributions (e.g., dietary, microbial, or methylation-related mechanisms) fall outside the scope of what the enrichment and pathway-membership databases directly support for this metabolite set and should be validated experimentally.